

High and fast carbon dioxide capture of hydroxypyridine-based ionogel depending on pore structure of mesoporous silica vesicle in the simulated flue gas

Chunfeng Xue^{a,*}, Lin Feng^a, Qian Zhang^a, Ruichao Huang^a, Yanan Hao^a, Xuli Ma^a, Gang Liu^a, Kaixi Li^b, Xiaogang Hao^{a,*}

^a Department of Chemical Engineering, Taiyuan University of Technology, Taiyuan, 030024, Shanxi, PR China

^b Institute of Coal Chemistry, Chinese Academy of Sciences, Taiyuan, 030001, PR China

ARTICLE INFO

Keywords:

Ionic liquid
Ionogel
Pore structure
CO₂ capture
Silica vesicle

ABSTRACT

Multiple active sites containing ionic liquid (IL) tetrabutylphosphonium 2-hydroxypyridine ([P₄₄₄₄][2-Op]) is prepared and loaded into mesoporous silica vesicle (MSV) with three kinds of pore structure for fabricating ionogels to capture CO₂ from simulated flue gas (CO₂/N₂). The ionogel MSV100-10 achieves CO₂ adsorption capacity of 1.69 mmol(g-ionogel)⁻¹ (that is 11.77 mmol(g-IL)⁻¹) in less than 10.0 min at 50 °C and CO₂ partial pressure of 0.2. The high adsorption capacity and rate can be ascribed to its large pore size and pore volume, which offer not only enough nanospace for highly dispersed IL but also necessary pathway for fast diffusion of guest molecule CO₂. Small mesopores of the supports benefit to form monomolecular layers for the lowly loaded IL due to their strong nano-confinement effect but are easily blocked by highly loaded IL. The support with large vesicle-like pores allow the formation of IL pairs for the lowly loaded IL due to their weak nano-confinement effect and keep necessary pathway for CO₂ diffusion in the highly loaded ionogel. The ionogel MSV100-10 performs capacity retention of 95% after 10 sorption cycles. The ionogel MSV100-10 featuring with low loading and cost, rapid adsorption, high capacity and excellent cyclic stability make it a competitive candidate in CO₂ capture from flue gas.

1. Introduction

Accumulated CO₂ as a kind of greenhouse gas in flue gas generated by the burning of fossil fuels especially from power plant resulted in direct threat to nature environment and human health (Cox et al., 2000). In view of this, more and more effort had been devoted to develop economic technology to eliminate the threat (Chu, 2009; Wang et al., 2014; Li et al., 2012; Cooper, 2015). The emissions reduction of CO₂ was on the agenda of many governments and regarded as one of the grim challenge for the coming centuries (Haszeldine, 2009). CO₂ capture was regarded as one of the essential technologies for recycling CO₂ and product purification (Chu, 2009; Cooper, 2015; Haszeldine, 2009).

Chemical absorption of CO₂ in amine-based solution was a common technology (Rochelle, 2009), but its industrial-scale implementation was still limited by serious equipment corrosion, easy solvent loss and high energy consumption during the absorbent regeneration (Spigarelli and Kawatra, 2013; Boot-Handford et al., 2014). Polyethyleneimine was also used to capture CO₂ from humid atmosphere despite of its high toxicity (Liu et al., 2015; Li et al., 2014). The ionic liquid (IL) for post-combustion

CO₂ capture was a good option to circumvent the aforementioned drawbacks due to its extremely low vapor pressure, and tunable property (Zhang et al., 2006; Brennecke and Gurkan, 2010; Zhang et al., 2011; Ramdin et al., 2012). Amine, amino acid, azolate, phenolate-based ILs had been developed to capture CO₂ (Wang et al., 2013; Gurkan et al., 2010; Wang et al., 2012, 2010). In view of their low CO₂ adsorption capacity, ILs with dual active sites were developed to achieve higher capacity (Zhang et al., 2013; Luo et al., 2014). Recently, pyridinolite IL showed CO₂ adsorption capacity of 10 wt% (1.28 mol·(mol-IL)⁻¹) (Luo et al., 2014). However, it had to take more than one hour to achieve saturation capacity in pure CO₂ due to the dramatically ascended viscosity of the IL during adsorption process (Cui et al., 2016).

Currently, loading the IL into the substrate with high specific surface area is a good solution to overcome the main limitations in practical applications represented by high viscosity besides high market cost. Synergistic effect of the formed ionogels showed improved CO₂ adsorption efficiency based on their enlarged gas-liquid interface (Tome and Marrucho, 2016; Wan et al., 2014; Horowitz and Panzer, 2014; Cheng et al., 2016). For example, IL [P₆₆₆₁₄][2-Op] (Trihexyl(tetradecyl)

* Corresponding authors.

E-mail addresses: cfxue@fudan.edu.cn (C. Xue), xghao@tyut.edu.cn (X. Hao).

<https://doi.org/10.1016/j.ijggc.2019.03.017>

Received 23 October 2018; Received in revised form 27 February 2019; Accepted 16 March 2019

Available online 27 March 2019

1750-5836/ © 2019 Elsevier Ltd. All rights reserved.

phosphonium 2-hydroxypyridine) infused in molecular sieve MCM-41 showed faster adsorption rate but lower CO₂ adsorption capacity than the pure IL (Cheng et al., 2016). Amino acid functionalized IL immobilized in SBA-15 displayed adsorption capacity of 0.61 mmol(g-ionogel)⁻¹ at room temperature in pure CO₂ (Hiremath et al., 2016). Similarly, IL [P₆₆₆₁₄] [Triz] (Trihexyl(tetradecyl)phosphonium 1,2,4-triazole) up to 50% impregnated in mesoporous silica SBA-15 displayed adsorption capacity of 0.675 mmol(g-ionogel)⁻¹ in pure CO₂ (Cheng et al., 2018). Their relatively low adsorption may be ascribed to aggregated IL depending on the pore structure of used substrate and poor accessibility related to the high loading. The long cylindrical mesopores also imposed diffusion limitation and resulted in low adsorption rate (Cheng et al., 2018). Moreover, the excessively loaded IL unavoidably led to inessential cost in the industrial application. The structures of supports affected the dispersal of IL and CO₂ diffusion, which were related to CO₂ absorption.

Therefore, silica with different pore structure was selected as the support. In this regard, the beneficial effect of using pore-expanded mesoporous silica supports had been demonstrated by several groups (Drese et al., 2009; Qi et al., 2012; Zhang et al., 2014; Yan et al., 2011; Wang et al., 2015). For example, the Sayari and Calleja groups had carried out extensive research about the adsorption of CO₂ by loading organic amines on conventional silica and pore-expanded HMS (Sanz-Perez et al., 2018), and MCM-41 (Sayari et al., 2016). The results indicated that their large pore size and pore volume provided huge space for organic amine and CO₂. The increased sorption capacity from 2.51 to 4.04 mol CO₂ kg⁻¹ was also reported by increasing the pore volume from 1.51 to 1.82 cm³ g⁻¹ of polyethyleneimine-impregnated silica at 75 °C and 0.15 bar (Yan et al., 2011). Moreover, the CO₂ sorption capacity of tetraethylenepentamine-impregnated ordered mesoporous silica was improved from 2.31 to 3.1 mol·kg⁻¹ by changing the pore size from 5.6 to 7.6 nm (Wang et al., 2015). It should be noted that the specific surface area, intrinsic pore structure, pore volume and shape played important roles during applications. The cylindrical pore was easily blocked at pore entrance by the IL. But vesicle-like pore can display better performance than the former in inhibiting the blocking.

In this paper, mesoporous silica vesicle (MSV) with different pore size is prepared as the support for loading pyridine-containing IL [P₄₄₄₄][2-Op] with small free volume. The IL displays a relatively low viscosity and two electronegative centers deriving from nitrogen and oxygen atoms (An et al., 2017). All the features benefit in getting high CO₂ adsorption capacity and rate. In view of its low viscosity, it is lowly loaded but highly dispersed in the support MSV to uptake CO₂ from the simulated flue gas at different temperature and partial pressure.

2. Experimental

2.1. Chemicals

Fumed silica (SiO₂), Trimethylbenzene (TMB), cetyltrimethyl ammonium bromide (CTAB), sodium hydroxide (NaOH), and anhydrous ethanol (EtOH) were purchased from Tianjin Zhiyuan Chemical Reagent Co, Ltd. Tetrabutylphosphonium hydroxide solution ([P₄₄₄₄][OH], 40% in H₂O) and 2-hydroxypyridine (2-Op) were purchased on J & K Scientific Ltd. Nitrogen (N₂) (> 99.99%), and carbon dioxide (CO₂) (> 99.99%) were bought from Taiyuan Taineng Gas Co. Ltd.

2.2. Preparation of pyridine-containing anion functionalized IL [P₄₄₄₄][2-Op]

The IL [P₄₄₄₄][2-Op] (tetrabutylphosphonium 2-hydroxypyridine) was prepared as followings: 19.5488 g [P₄₄₄₄][OH] was dissolved in 40 mL EtOH and stirred at room temperature. Then 2.690 g 2-Op was dissolved in another 40 mL EtOH. Subsequently, the two solutions were mixed at 30.0 °C for 24.0 h and evaporated in a rotary evaporator at 50.0 °C for 1.0 h. After drying the mixture at 60.0 °C under vacuum for 48.0 h, we finally obtained the IL. The CO₂ adsorption capacity of pure

IL was measured as 1.60 mmol·g⁻¹ by bubbling pure CO₂ at 50 °C for more than 100 min.

2.3. Preparation of mesoporous silica vesicle (MSV)

The molar ratio of raw materials was selected as followings: 100.0 SiO₂: w TMB: 25.0 NaOH: 10.0 CTAB: 2000.0 H₂O (w = 50, 75, 100). Typically, 2.4540 g CTAB was firstly dissolved and stirred in 24.000 g H₂O. Then 0.6680 g NaOH and 10.4156 g TMB were added with further stirring at room temperature to get a clear solution. Finally, 4.0000 g fumed SiO₂ was added with strong stirring for 0.50 h. The obtained mixture was transferred into autoclave and heated at 70.0 °C for 8.0 h. The white solid was washed with distilled water. The washed solid was dried at 115.0 °C for 12.0 h and further calcined at 550.0 °C for 8.0 h in muffle oven to remove the used CTAB and TMB. The ramp rate of 2.0 °C min⁻¹ is used. The obtained samples were denoted as MSV50, MSV75, and MSV100, respectively.

2.4. Preparation of ionogels

The ionogels were prepared as followings: A given amount of dried IL was dissolved in 30.0000 g ethanol and stirred for 1.0 h at room temperature. Then 1.0000 g support was added into the solution and mixed at 80.0 °C for 2.0 h. The resultant mixture was evaporated at 80.0 °C for 3.0 h and dried in a vacuum oven at 100.0 °C for 12.0 h to remove the residual ethanol. According to the IL [P₄₄₄₄][2-Op] percentage of 2.0%, 5.0%, 10.0%, 15.0%, 30.0% loaded into the support MSV, the obtained ionogels were denoted as MSV50-2, MSV50-5, MSV50-10, MSV50-15, MSV50-30, MSV75-2, MSV75-5, MSV75-10, MSV75-15, MSV100-30, MSV100-2, MSV100-5, MSV100-10, MSV100-15, and MSV100-30, respectively.

2.5. Characterization

X-ray diffraction (XRD) patterns of samples were collected on a Rigaku Ultima IV type X-ray diffractometer in a range of 2θ = 1.0–7.0°, equipped with Cu K_α radiation (λ = 0.154056 nm, 40.0 kV, and 40.0 mA). Transmission Electron Microscope (TEM) was carried out on a JEOL JEM 3010 operated at an acceleration voltage of 300.0 kV. Fourier transfer infrared spectra (FT-IR) were recorded with a solution of 2.0 cm⁻¹ in the range of 400.0–4000.0 cm⁻¹ on a Thermo Nicolet-360 FT-IR spectrometer using KBr as the window material. N₂ adsorption–desorption isotherm was measured on a JW-BK122 W specific surface area and pore size analyzer (Beijing JWGB Sci. & Tech. Co., Ltd.) at -196.0 °C. The support MSV (100.0–200.0 mg) was degassed at 300.0 °C for 3.0 h in vacuum before the measurement, while the ionogel (100.0–200.0 mg) was degassed at 120.0 °C for 8.0 h. Specific surface area was calculated using Brunauer–Emmett–Teller (BET) method. Pore size distribution was calculated from the adsorption branches of the isotherms using the Barrett–Joyner–Halenda (BJH) model. The water content of the sample was measured using an AKF-3 Karl Fischer coulometer. Thermogravimetry (TG) curve was collected on a Netzsch STA 449 F5 TG/DTG instrument. About 7.0 mg sample was placed onto an alumina crucible and heated at a rate of 10.0 °C min⁻¹ from 30.0 to 800.0 °C in N₂ flow (100 mL min⁻¹) to analyze thermal stability of the ionogel. ¹³C, ³¹P, and ¹H NMR spectrum of 0.50 mL of IL [P₄₄₄₄][2-Op] were recorded on a Bruker DR X 500 MHz. ¹³C NMR (500.0 MHz, DMSO, 25.0 °C): δ = 13.3, 14.1, 18.6, 18.9, 23.6, 23.7, 23.8, 23.9, 24.3, 27.3, 27.8, 105.5, 120.1, 132.2, 139.7, 160.4 ppm (Fig. 1(a)); ³¹P NMR (500.0 MHz, DMSO, 25.0 °C) of the IL: δ = 33.8 ppm (Fig. 1(b)); (c) ¹H NMR (500.0 MHz, DMSO, 25.0 °C) of IL [P₄₄₄₄][2-Op]: δ = 0.90, 1.36–1.51, 2.16–2.25, 5.71, 6.86 and 7.56 ppm (Fig. 1(c)).

2.6. Studies on CO₂ adsorption in simulated flue gas

CO₂ adsorption capacities of the ionogels were calculated by recording breakthrough curves on a Gasboard-3100 gas analyzer. 1.0 g ionogel was placed into the sample tube with an inner diameter of

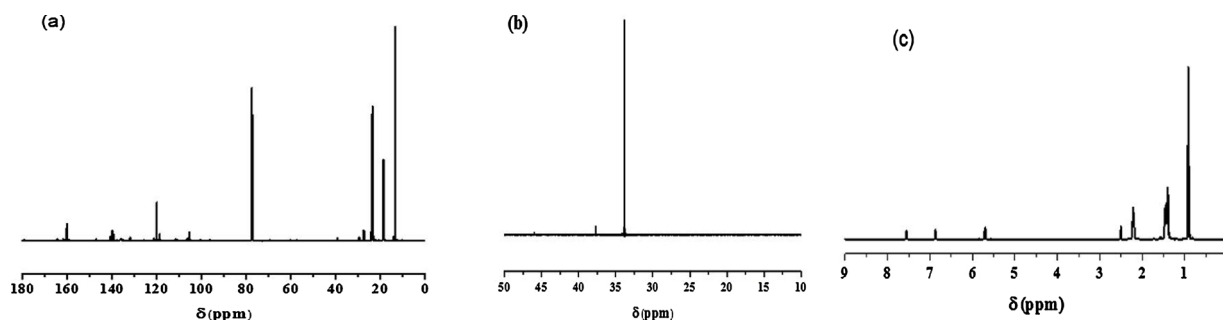


Fig. 1. (a) ^{13}C , (b) ^{31}P , and (c) ^1H NMR spectrum of IL $[\text{P}_{4444}][2\text{-Op}]$.

10.0 mm and a length of 220.0 mm. First, the sample was treated at 120.0°C for 2.0 h in N_2 flow ($98.5\text{ mL}\cdot\text{min}^{-1}$) to remove trace water and adsorbed gas impurity. Then it was cooled to a given adsorption temperature, which referred to the typical temperature range of flue gas. Next, a mixed CO_2/N_2 as simulated flue gas was introduced into the analyzer via a bypass until the CO_2 concentration was constant, in which the volume concentration of CO_2 was approximately 15.0 vol.% in N_2 . Afterwards, the simulated flue gas was switched to the sample tube at 50.0°C to perform the adsorption experiment. The experiment was completed when the CO_2 analyzer showed a constant CO_2 concentration that was equal to the initial one. The CO_2 adsorption capacity (q , $\text{mmol}\cdot(\text{g}\cdot\text{ionogel})^{-1}$) of the sample at a certain time (t , s) was calculated from the equation below:

$$q = \int_0^t Q(c_{\text{in}} - c_{\text{eff}}) dt / (mv_m)$$

Where m (g) represents the mass of the sample. Q ($\text{mL}\cdot\text{min}^{-1}$) is the influent flow rate. c_{in} (vol.%) and c_{eff} (vol.%) are the influent and effluent CO_2 concentration, respectively. The v_m ($22.4\text{ mL}\cdot\text{mmol}^{-1}$) is standard molar volume. Herein, ionogel MSV100-10 was chosen to investigate the cyclic stability in 10 cycles of adsorption and regeneration. The process of regeneration was conducted in pure N_2 flow at $98.5\text{ mL}\cdot\text{min}^{-1}$ at 120.0°C for 1.0 h. The adsorption capacity of MSV100-10 was also evaluated at partial pressures of 0.20, 0.15, and 0.10 as well as temperatures of 30, 50, and 70°C .

3. Results and discussion

3.1. XRD patterns of the supports and ionogels

The XRD patterns of the supports are shown in Fig. 2. The sample MSV50 shows a strong but broad diffraction peak at $2\theta = 1.85^\circ$ (Fig. 2A(a)), implying a less ordered mesostructure (Kresge et al., 1992; Wan and Zhao, 2007; Zhao et al., 1998; Xue et al., 2008). With the addition of TMB, a broad but weak peak is observed at the similar position for the obtained sample MSV75 (Fig. 2B(a)). The results imply further decreasing of species with ordered mesostructure. No obvious peak can be observed for the sample MSV100 (Fig. 2C(a)), which is prepared from the system containing excessive TMB. All the results confirm that the introduction of more and more TMB seriously disturbs the behavior of the template CTAB and is not favorable for assembling ordered mesostructure.

With more and more IL $[\text{P}_{4444}][2\text{-Op}]$ loaded into the support MSV-50, the characteristic diffraction peaks of the obtained ionogels gradually weaken even to disappear (Fig. 2A(b–f)). The decaying intensity of the peak implies that the IL is surely impregnated into the mesopores, which weaken the contrast of diffraction patterns (Sanz-Perez et al., 2013). In case of the ionogels based on the supports MSV75 and MSV100, all of them display disordered mesostructure similar to the supports themselves (Fig. 2B(b–f) and 2 C(b–f)). In all, whether the loading percentage is low as 2% or high as 30%, the IL is well infused or lined into the support.

3.2. TEM images of the supports

From TEM images of supports MSV50, MSV75, and MSV-100, it is found that their irregular morphology and different textural property resulting from the addition of more and more TMB (Fig. 3). They do not show conventional columnar pores. Combining with other literatures (Zhang et al., 2007; Grosso et al., 2007; Yang et al., 2018), vesicle-like pores can be concluded from their images. In detail, the support MSV50 shows many but less ordered pores (Fig. 3A, B), coinciding with its XRD result (Fig. 2a). The presented domains disclose that the vesicle-like pore is uniform and the diameter is about 3.8 nm (Fig. 3C) (Jiang et al., 2007). The support MSV75 shows a mixture of less ordered and disordered texture (Fig. 3D), agreeing with the result concluded from its decaying diffraction peak (Fig. 2b). Its less ordered structure is similar to that of the support MSV50. However, the pore diameter of about 3.4 nm is smaller than that of MSV50 (Fig. 3F). Its large vesicle-like pore is obviously disordered and the diameter is about 11.0 nm (Fig. 3E), which is also different from MSV50. The results indicate the occurrence of phase separation during the mesostructure assembly due to the addition of more TMB. In case of the support MSV100, the large vesicle-like pores are completely disordered but uniform (Fig. 3H, I), coinciding with its disappeared XRD peak (Fig. 2c). Their diameters are centered at about 10.0 nm (Fig. 3J). All the results reveal that the mesostructure transformation process is accompanied with enlarged pores when the TMB is continuously introduced in the system.

3.3. Pore property of the supports and ionogels

N_2 adsorption isotherms demonstrate that all the supports exhibit type IV curves with obvious hysteresis loops but at different relative pressures, implying the presence of mesopores with different diameters (Wan et al., 2014). In details, the sample MSV50 shows a small hysteresis loop at a low P/P_0 between 0.3 and 0.5 (Fig. 4A(a)), implying the presence of small mesopores. Corresponding pore size distribution reveals its pore diameter centered at 3.60 nm (Fig. 4B(a)), agreeing with the above TEM observation. Its specific surface area and pore volume is $1033\text{ m}^2\text{ g}^{-1}$ and $1.25\text{ cm}^3\text{ g}^{-1}$ (Table 1), respectively. The sample MSV75 exhibits a large loop at $P/P_0 = 0.40\text{--}0.85$ (Fig. 4C(a)), implying the existence of hierarchical pores. Correspondingly, its pore diameters mainly center at 3.4 and 10.7 nm (Fig. 4D(a)), also confirming the TEM results. It displays a smaller specific surface area of $1016\text{ m}^2\text{ g}^{-1}$ but a larger pore volume of $1.66\text{ cm}^3\text{ g}^{-1}$ than MSV50. A large but narrow loop at $P/P_0 = 0.40\text{--}0.90$ is observed for the sample MSV100 (Fig. 4E(a)), implying the appearance of large mesopores. Its pore diameter mainly centers at 10.3 nm (Fig. 4F(a)). It shows a largest pore volume of $2.13\text{ cm}^3\text{ g}^{-1}$ and smallest specific surface area of $987\text{ m}^2\text{ g}^{-1}$ among the supports.

The incorporations of ILs also result in a ripple effect on textural characteristics of the resultant ionogels, especially on their specific surface area, pore volume, and pore size. With increasingly infused IL $[\text{P}_{4444}][2\text{-Op}]$ in the support MSV50, N_2 adsorption isotherms of the resultant ionogels still show type IV curves with hysteresis loops at different relative pressures (Fig. 4A(b–f)), indicating their evolution of pore structure. The specific surface area continuously decreases from $936\text{ m}^2\text{ g}^{-1}$ for

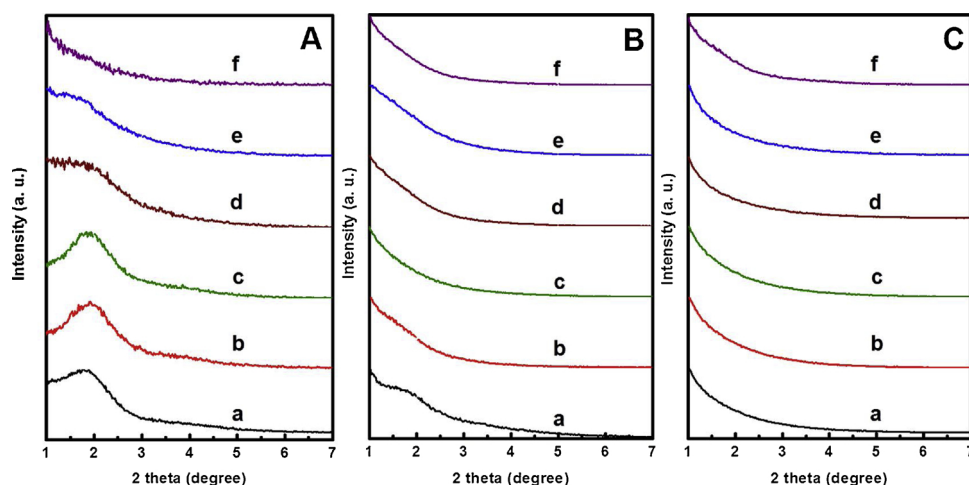


Fig. 2. Low-angle XRD pattern of samples: (A) (a) MSV50, (b) MSV50-2, (c) MSV50-5, (d) MSV50-10, (e) MSV50-15, and (f) MSV50-30; (B) (a) MSV75 (b) MSV75-2, (c) MSV75-5, (d) MSV75-10, (e) MSV75-15, and (f) MSV75-30; (C) (a) MSV100 (b) MSV100-2, (c) MSV100-5, (d) MSV100-10, (e) MSV100-15, and (f) MSV100-30.

ionogel MSV50-2 to $486 \text{ m}^2 \text{ g}^{-1}$ for ionogel MSV50-30. Obviously, the hysteresis loops shift to lower pressure and become slim in turn, implying reduced pore size and volume. In detail, the pore diameter shrinks from 3.2 nm for the ionogel MSV50-2 to 2.4 nm for the ionogel MSV50-30 (Fig. 4B(b)–(f)). Correspondingly, its pore volume evolves from 1.19 to $0.73 \text{ cm}^3 \text{ g}^{-1}$ for the final ionogel MSV50-30. All the results indicate that the IL is successfully lined in pore surface of the support MSV50. The remained pores can act as necessary pathway for CO_2 diffusion. In case of the ionogels based on the support MSV75, they still show type IV curves with broad hysteresis loops (Fig. 4C(b)–(f)), implying the presence of dual-model pores similar to the support itself. The specific surface area of the ionogels gradually decreases from $956 \text{ m}^2 \text{ g}^{-1}$ with the impregnated IL progressively increased from 2.0%–30.0%. According to

their pore size distributions (Fig. 4D(b)–(f)), both of two main pore diameters shift from 3.4 and 10.6 nm to 2.2 and 8.2 nm. The results imply that the IL is uniformly lined in the inner surface of both the small and large mesopores. As a result, their pore volumes turn from $1.59 \text{ cm}^3 \text{ g}^{-1}$ of ionogel MSV75-2 to $0.98 \text{ cm}^3 \text{ g}^{-1}$ of MSV75-30. In case of the ionogels based on the support MSV100, they show type IV curves with obvious hysteresis loops at high $P/P_0 = 0.50$ – 0.90 (Fig. 4E(b)–(f)), implying the existence of large mesopores similar to the support itself. Also, they show low and low N_2 adsorption capacities, reflecting the extent of pore filling. The pore volume turns from $1.96 \text{ cm}^3 \text{ g}^{-1}$ of MSV100-2 to $0.96 \text{ cm}^3 \text{ g}^{-1}$ of MSV100-30. The pore diameter center shifts from 9.7 to 8.4 nm due to the progressive filling of pores with the IL according to the distributions (Fig. 4F(b)–(f)).

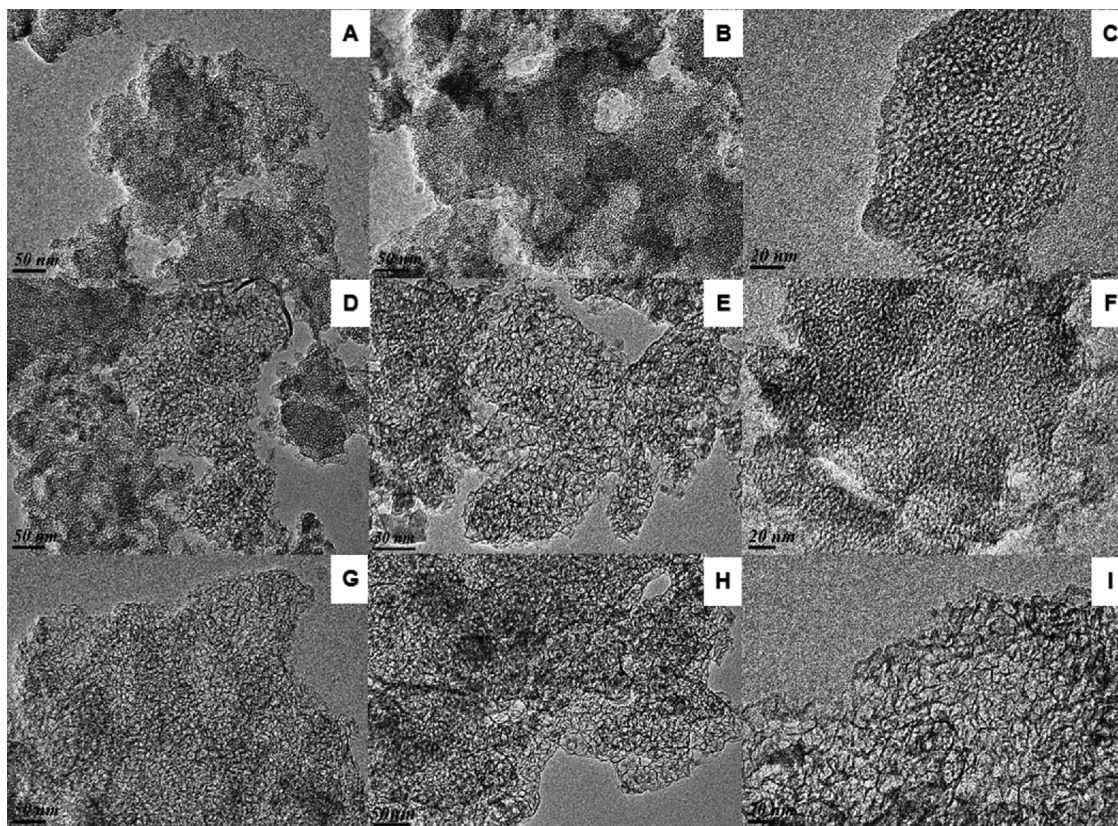


Fig. 3. TEM images of samples: (A, B, C) MSV50; (D, E, F) MSV75; and (H, I, J) MSV100.

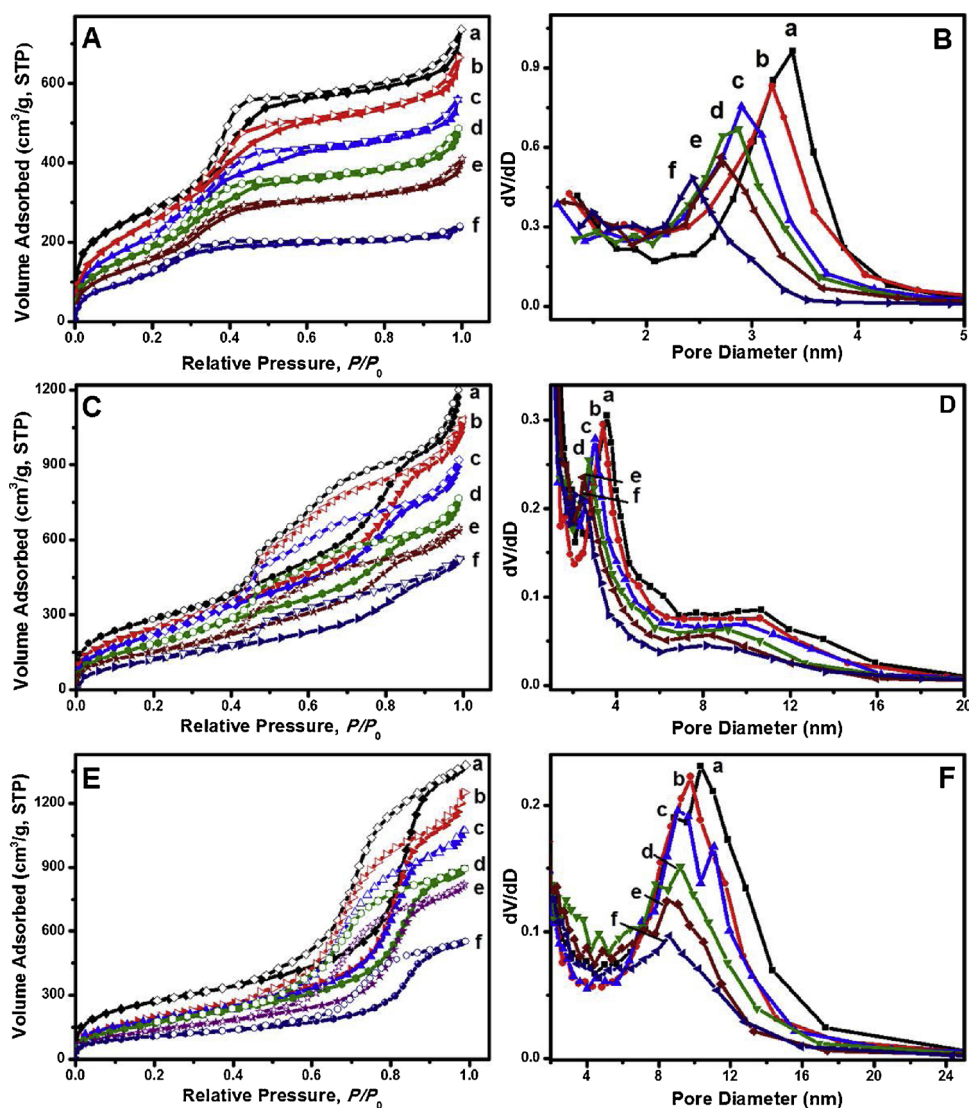


Fig. 4. N₂ adsorption-desorption isotherms (A) and pore size distributions (B) of support MSV-50 and ionogels based on it: (a) MSV50, (b) MSV50-2, (c) MSV50-5, (d) MSV50-10, (e) MSV50-15, and (f) MSV50-30; N₂ adsorption-desorption isotherms (C) and pore size distributions (D) of support MSV75 and ionogels based on it: (a) MSV75, (b) MSV75-2, (c) MSV75-5, (d) MSV75-10, (e) MSV75-15, and (f) MSV75-30; N₂ adsorption-desorption isotherms (E) and pore size distributions (F) of support MSV100 and ionogels based on it: (a) MSV100, (b) MSV100-2, (c) MSV100-5, (d) MSV100-10, (e) MSV100-15, and (f) MSV100-30.

Table 1

Texture property, and CO₂ adsorption capacity in simulated flue gas of samples.

Sample	S _{BET} (m ² ·g ⁻¹)	Pore size (nm)	Pore volume (cm ³ ·g ⁻¹)	CO ₂ adsorption capacity at 50 °C (mmol·(g-ionogel) ⁻¹)
MSV50	1033	3.6	1.25	0.35
MSV75	1016	3.6, 10.7	1.66	0.45
MSV100	987	10.3	2.13	0.57
MSV50-2	936	3.2	1.19	0.66
MSV50-5	891	3.1	1.02	0.79
MSV50-10	775	2.9	0.84	1.05
MSV50-15	651	2.7	0.73	0.52
MSV50-30	486	2.4	0.67	0.29
MSV75-2	956	3.4, 10.6	1.59	0.88
MSV75-5	813	2.9, 9.9	1.37	1.02
MSV75-10	718	2.8, 9.1	1.16	1.13
MSV75-15	604	2.5, 8.4	1.09	0.50
MSV75-30	499	2.2, 8.2	0.98	0.30
MSV100-2	936	9.7	1.96	0.96
MSV100-5	878	9.3	1.82	1.14
MSV100-10	847	9.2	1.38	1.49
MSV100-15	749	8.5	1.15	1.30
MSV100-30	572	8.4	0.96	0.87

3.4. FT-IR spectra of the IL, support, and ionogel

The structure of IL [P₄₄₄₄][2-Op] is also confirmed by comparing its FTIR pattern with those of the raw materials. The bands at around 2927 cm⁻¹ ascribed to the stretching vibration of -CH₃ group are observed for raw material [P₄₄₄₄][OH] and IL [P₄₄₄₄][2-Op] (Fig. 5A(a), Fig. 5B(a), and 5 A(c)) (Mello et al., 2011). The bands at 1601 and 1501 cm⁻¹ for the N-H bending vibration are observed in the spectra of 2-Op and IL [P₄₄₄₄][2-Op]. Combining these with the above NMR results, it is concluded that the IL is well prepared. For the support MSV100, the broad band at 3415 cm⁻¹ is assigned to the stretching vibration of Si-OH and physically adsorbed H₂O (Fig. 5B(b)) (Johari et al., 2014). The bands at 1079, 791, and 455 cm⁻¹ are attributed to the asymmetric and symmetric stretching vibrations of Si-O-Si and the bending vibration of Si-O, respectively (Fig. 5B(b)) (Jiang et al., 2007). All the bands mentioned above can be also observed in the ionogels MSV100-2, MSV100-10, and MSV100-30 (Fig. 5B(c)-(e)). Reasonably, their relative intensity of the characteristic bands derived from the IL gradually boosts as the loaded IL percentage is increased from 2% to 10%, and 30% (Fig. 5B(c)-(e)).

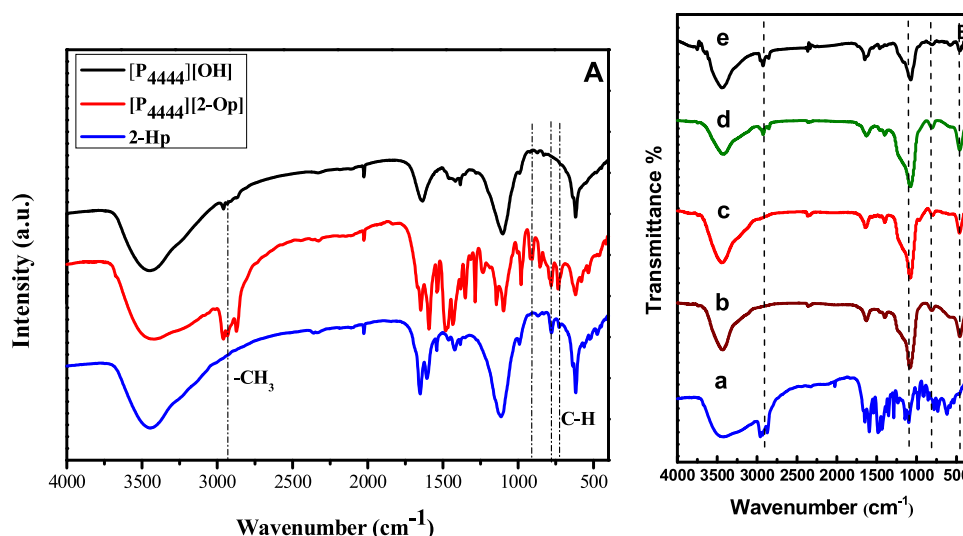


Fig. 5. FT-IR spectra of samples: (A) IL [P₄₄₄₄][2-Op] along with its synthetic reactants; (B). (a) IL [P₄₄₄₄][2-Op]; (b) MSV100; (c) MSV100-2; (d) MSV100-10; and (e) MSV100-30.

3.5. Loading-dependent thermal stability of the ionogels

To evaluate thermal stability of ionogels as well as the IL, their TG curves are collected and piled for careful investigation (Fig. 6). The support MSV50 only shows a mini weight loss of 1.0%, which can be ascribed to the removal of hydroxyl group at high temperature (Fig. 6A(a)). The results imply that it is well activated for subsequent IL infusion. In case of the pure IL, no obvious weight loss is observed below 150 °C in its curve (Fig. 6A(b)), implying an ignorable water content. An outstanding weight loss of 99% is observed in the range from 150 to 330 °C, which can be attributed to the decomposition of the organic species. Accordingly, its onset decomposition temperature is about 230 °C, indicating a better thermal stability than common liquid. Interestingly, the IL immobilized in the support MSV50 shows different thermal stability depending on the loading percentage. For the ionogel MSV50-2 with the IL loading percentage of 2.0%, its weight loss starts at 430 °C and mainly occurs around 530 °C (Fig. 6A(c) and 6 B(c)), which is obviously higher than the decomposition temperature of the bulk IL. The results indicate that the thermal stability of the nano-confined IL is dramatically enhanced by around 200 °C. It is also different from reported poor stability of other IL on the similar substrate (Singh et al., 2014). The enhancement on thermal stability can be attributed to the significant interfacial layering and stiffening of the loaded IL in the vicinity of the support surface resulting from the nano-confinement effect (Coasne et al., 2011). Combining with other above results including N₂ sorption isotherms, the results imply that the IL prefers to dot the pore surface in a monomolecular layer style so that showing high onset decomposition temperature, which can also fully flat and strongly interact with the pore wall as well as exposed more active sites for capture CO₂. No obvious weight loss is observed below 150 °C. A small weight loss occurring between 150 and 430 °C is observed, which covers the decomposition temperature of the bulk IL. The results imply that a small part of the loaded IL shows a similar thermal stability to the pure IL. With further impregnation of the IL up to 5.0% and 10.0%, the obtained ionogels MSV50-5 and MSV50-10 display two obvious weight losses in the two ranges of 230–430 °C and 430–630 °C (Fig. 6A(d)–(e) and 6 B(d)–(e)), corresponding to the decomposition of aggregating IL and monomolecule layering one, respectively. The results indicate that the IL may stay in two different statuses, which separately show a similar thermal stability to the bulk IL and an enhanced stability. What's more, the weight loss occurring at 230–430 °C gradually enhances and correspondingly the one occurring at 430–630 °C progressively decays. The results imply the transformation trend of the

status of the loaded IL with more and more IL impregnation. Also, when the loading percentage up to 30.0%, only the weight loss occurring at 230–430 °C can be observed for the ionogel MSV50-30 (Fig. 6A(g) and 6 B(g)), implying the complete appearance of the aggregating IL. The curves of the ionogels based on the support MSV75 show a similar evolution rhythm of weight losses occurring at different temperature ranges to those of ionogels based on MSV50 (Fig. 6C(c)–(g) and 6 D (c)–(g)). With the IL loading percentage from 2.0%, to 5.0%, 10.0%, 15.0%, and 30.0%, the status of the loaded IL transforms from pseudo-simplex monomolecular layer until to simplex aggregation by going through their coexistence. The results well confirm the feature of the IL dispersing status mainly depends on its loading percentage besides the pore structure. In case of the ionogels based on the support MSV100, their weight losses still exhibit the similar developing pace depending on the increased IL loading percentage (Fig. 6E(c)–(g) and 6 F(c)–(g)). Noticeably, a well-resolvable weight loss can be observed at 450 °C for the ionogels MSV100-5, MSV100-10 and MSV100-15 (Fig. 6F(d)–(f)), implying the presence of obvious IL pairs or oligomers different from the monomolecular layer aforementioned. Reasonably, the mesopore size or space of the support MSV100 is large enough to form IL pairs or oligomers and impose a weak nano-confinement effect. The thermal stability of the IL pairs or oligomers may fall in between that of bulk IL and that of the highly layering IL.

3.6. CO₂ adsorption in the ionogels

The CO₂ adsorption performances of ionogels basing on different supports with IL [P₄₄₄₄][2-Op] loading percentage ranging from 0.0–30.0% are measured at 50.0 °C and partial pressure of 0.15 (Fig. 7A–C). We also summarize their CO₂ adsorption capacities of three batches samples for further discussion (Fig. 8). The supports MSV50, MSV75, and MSV100 show adsorption capacity of 0.35, 0.45, and 0.57 mmol·(g-ionogel)⁻¹, respectively ((a) in Fig. 7A–C). The results can be ascribed to the accessibility of surface hydroxyl depending on their different pore size and pore volume. After the IL [P₄₄₄₄][2-Op] (2.0%–10.0%) is continuously impregnated in the support MSV50, CO₂ adsorption capacity of the ionogels MSV50-2, MSV50-5 and MSV50-10 increases enormously from 0.66 to 0.79, and 1.05 mmol·(g-ionogel)⁻¹ in the 7, 10, 14 min respectively (Fig. 7A(b)–(d)). The results indicate the lowly loaded IL in the ionogels show fairly faster and higher adsorption than the bulk IL. The highly dispersed IL in a monomolecular layer style may expose more active sites to capture more CO₂ and impose low diffusion limitation to achieve fast adsorption. When the IL [P₄₄₄₄][2-

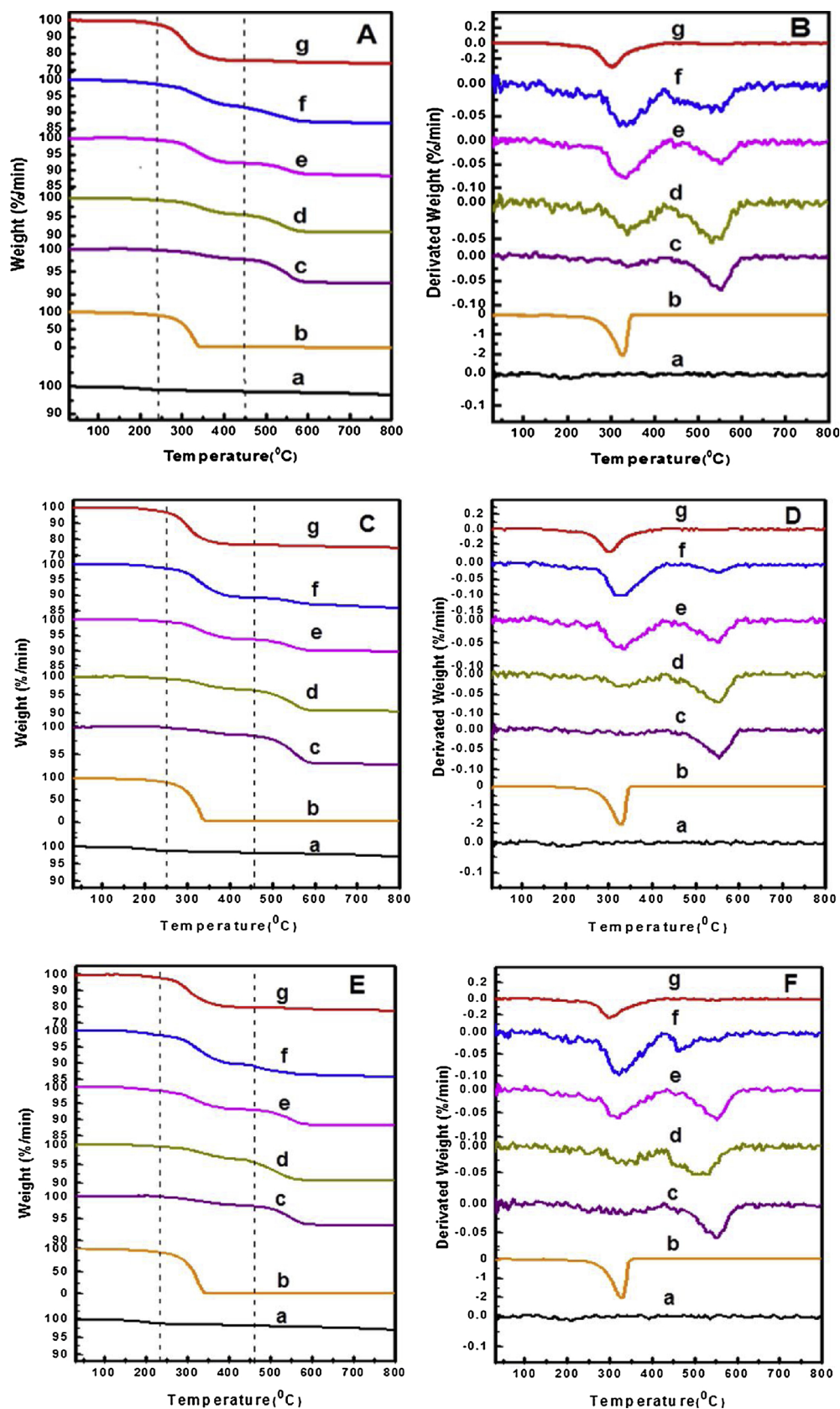


Fig. 6. TG (A) and DTG (B) curves of ionogels based on MSV50: (a) MSV50, (b) bulk IL, (c) MSV50-2, (d) MSV50-5, (e) MSV50-10, (f) MSV50-15, and (g) MSV50-30; TG (C) and DTG (D) curves of ionogels based on MSV75: (a) MSV75, (b) bulk IL, (c) MSV75-2, (d) MSV75-5, (e) MSV75-10, (f) MSV75-15, and (g) MSV75-30; TG (E) and DTG (F) curves of ionogels based on MSV100: (a) MSV100, (b) bulk IL, (c) MSV100-2, (d) MSV100-5, (e) MSV100-10, (f) MSV100-15, and (g) MSV100-30.

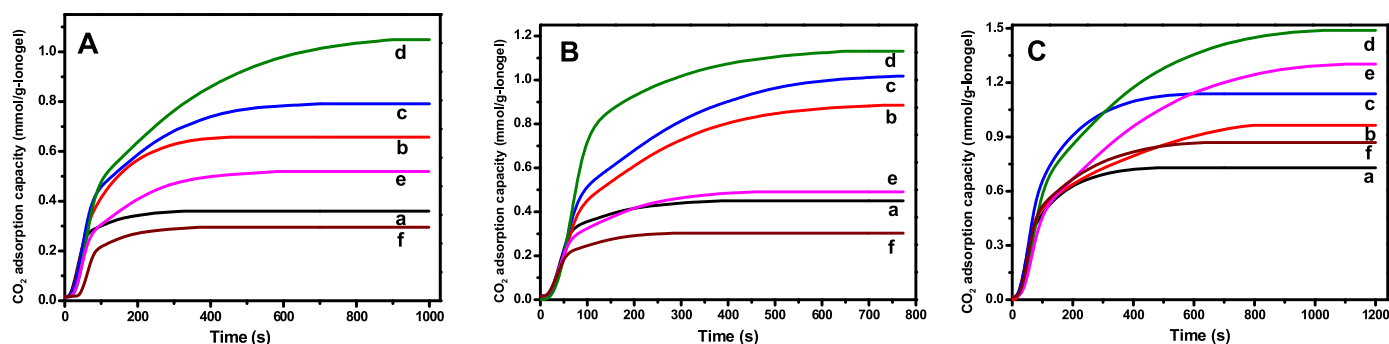


Fig. 7. CO₂ adsorption curves at 50 °C of the supports and ionogels: (A) (a) MSV50, (b) MSV50-2, (c) MSV50-5, (d) MSV50-10, (e) MSV50-15, and (f) MSV50-30; (B) (a) MSV75, (b) MSV75-2, (c) MSV75-5, (d) MSV75-10, (e) MSV75-15, and (f) MSV75-30; (C) (a) MSV100, (b) MSV100-2, (c) MSV100-5, (d) MSV100-10, (e) MSV100-15, and (f) MSV100-30.

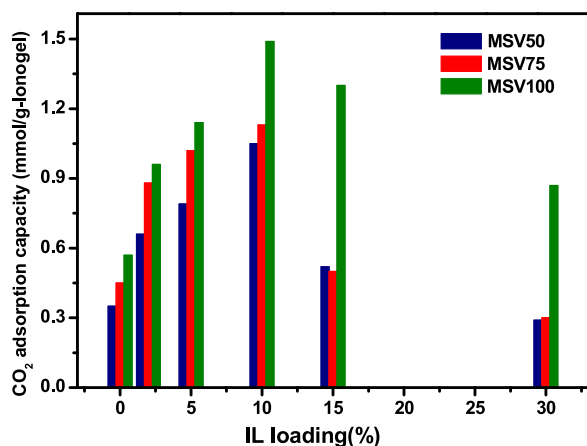


Fig. 8. CO₂ adsorption capacity at 50 °C from three batches of the supports MSV50, MSV75, and MSV100 and ionogels based on them with different IL loading percentage.

Op] is further impregnated up to 15.0% and 30.0%, the obtained ionogels MSV50-15 and MSV50-30 display declined CO₂ adsorption capacity of 0.52 and 0.29 mmol·(g-ionogel)⁻¹ (Fig. 7A(e)–(f)), respectively. The results can be ascribed to the formation of IL aggregation and pore blocking. In case of ionogels based on the support MSV75, their CO₂ adsorption capacities show the similar evolution trend to those of ionogels based on MSV50 (Fig. 7B(b)–(f)). However, the former ionogels show higher adsorption capacity than the latter when they are loaded with same percentage IL (Fig. 7A(b)–(d) and 7 B(b)–(d)). The large pore diameter and volume of the support MSV75 may be responsible for the feature, which benefits from its more exposed active sites at the same IL loading. Noticeably, the highly loaded ionogels MSV75-15 and MSV75-30 show CO₂ adsorption capacity close even lower than the blank support (Fig. 7B(e)–(f)), imply the lacking of diffusion pathway due to the possible pore blocking. The ionogels based on the support MSV100 show the highest adsorption capacity among them with the same IL loading percentage (Fig. 7C). It respectively takes about 13, 10, and 16 min for the ionogels MSV100-2, MSV100-5, and MSV100-10 to achieve the enhanced CO₂ saturation adsorption. Typically, the ionogel MSV100-10 shows the highest CO₂ adsorption capacity of 1.49 mmol·(g-ionogel)⁻¹ among all the ionogels (Fig. 7C(d)). The results can be ascribed to its more exposed active sites and CO₂ diffusion pathways in the large mesopore. Similarly, the highly loaded ionogel MSV100-30 shows lower CO₂ adsorption capacity than the lowly loaded ionogels including MSV100-2, MSV100-5, and MSV100-10 but higher than the blank support (Fig. 7C(e)–(f)), implying the blocking of part of diffusion pathway due to the highly loaded IL.

Compared to ionogels based on the other support with cylindrical mesopores such as SBA-15, the support with vesicle-like pores here still shows higher CO₂ adsorption capacity, implying that the support MSV100 is favorable for high dispersing and difficult to be blocked even at high IL loading. (Cheng et al., 2018)

3.7. Influences of operation temperature and partial pressure on CO₂ adsorption capacity

In practical applications, the operation temperature sometimes fluctuates in the flue gas. In view of this, the influence of temperature on CO₂ adsorption capacity is investigated at partial pressure of 0.15. Fig. 9A depicts CO₂ saturation adsorption capacity for the ionogels MSV50-10, MSV75-10, and MSV100-10 at 30, 50, and 70 °C. They are 1.11, 1.26, and 1.59 mmol·(g-ionogel)⁻¹ at 30 °C and higher than 0.8, 0.9, and 1.22 mmol·(g-ionogel)⁻¹ at 70 °C in the simulated flue gas. The described trend may be related to the exothermic nature of both IL-related physisorption and chemisorption processes. At the same time, the results suggest that the interaction between CO₂ and the loaded IL is moderate and benefits to the regeneration at a cost-effective temperature (Hiremath et al., 2016).

For future applications, CO₂ adsorption performances of the ionogel MSV100-10 at various partial pressures of 0.20, 0.15 and 0.10 are investigated at 50 °C (Fig. 9B). It takes less than 10 min to achieve the enhanced adsorption capacity of 1.69 mmol·(g-ionogel)⁻¹ for the ionogel MSV100-10 at the CO₂ partial pressure of 0.20. As expected, it keeps the same CO₂ adsorption capacity of around 1.49 mmol·(g-ionogel)⁻¹ as mentioned above. It is reasonable that long time is spent in attaining the saturation adsorption at the low partial pressure of 0.15. Its CO₂ adsorption capacity further decreased to 1.17 mmol·(g-ionogel)⁻¹ after balancing at the lower partial pressure of 0.10 in about 18 min. Generally, both adsorption capacity and rate go down with the decreasing CO₂ partial pressure.

3.8. Cyclic stability of CO₂ adsorption

The attractive cyclic stability of the ionogel is often more important than high adsorption capacity in industrial applications. The ionogel MSV100-10 is selected to perform ten adsorption cycles to evaluate the cyclic stability. Its CO₂ adsorption capacity in each cycle is also summarized (Fig. 10). It can be seen that the adsorption capacity slightly fluctuates from 1.49 to 1.42 mmol·(g-ionogel)⁻¹ at the final cycle. The capacity retention is more than 95%, imply an excellent cyclic stability. The attractive performance may be related to the modified properties such as strong interfacial layering and low dynamics. The good cyclic stability and high adsorption capacity make the low-loaded ionogel a good candidate for capture CO₂ from the flue gas.

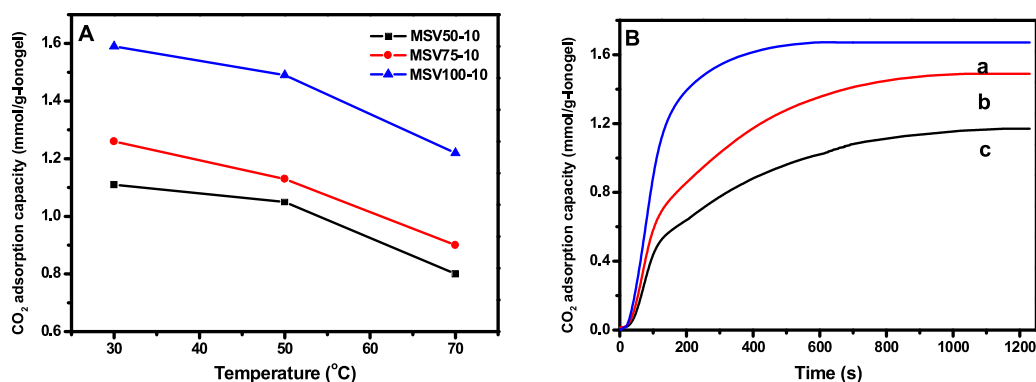


Fig. 9. (A) CO₂ saturation adsorption capacity at different temperatures of ionogels; (B) CO₂ adsorption curves of the ionogel MSV100-10 recorded at 50 °C in simulated flue gas with CO₂ partial pressures: (a) 0.20, (b) 0.15, and (c) 0.10.

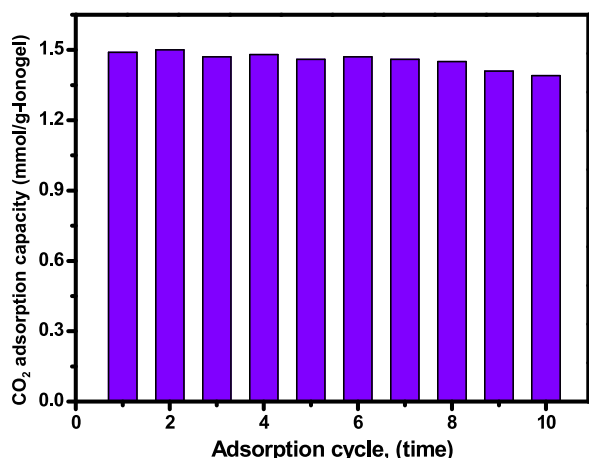


Fig. 10. Cyclic stability of ionogel MSV100-10 recorded at 50 °C and partial pressure of 0.15.

4. Conclusions

Three kinds of mesostructured silica vesicle is prepared and used to load multiple active sites containing ionic liquid [P₄₄₄₄][2-Op] for fabricating ionogels to capture CO₂ from the simulated flue gas (CO₂/N₂). Besides the IL loading level, the dispersing state of the loaded IL mainly depends on pore diameter as well as pore volume of the support and finally determines the adsorption capacity and rate. Small mesopores of the support benefit to form the monomolecular layer for the lowly loaded IL due to their strong nano-confinement effect but are easily blocked by the highly loaded IL. The vesicle-like support with large pore size and volume has priority to load as much highly dispersed IL as possible and simultaneously keep smooth highway for guest molecule diffusion. The ionogel MSV100-10 can achieve CO₂ adsorption capacity of 1.69 mmol(g-ionogel)⁻¹ (that is 11.77 mmol(g-IL)⁻¹) in less than 10 min at 50 °C and partial pressure of 0.20. It performs capacity retention of 95% after 10 sorption cycles. The features of low loading and cost, rapid adsorption, high capacity and excellent cyclic stability make it a competitive candidate in CO₂ capture from the flue gas.

Acknowledgements

The authors acknowledge the financial supports from China Scholarship Council (No. 201506935028), Key Scientific and Technological Projects of Shanxi Province (No. MD2014-09), and Natural Science Foundation of Shanxi Province (No. 201801D121060).

References

- An, X., Du, X., Duan, D., Shi, L., Hao, X., Lu, H., Guan, G., Peng, C., 2017. An absorption mechanism and polarity-induced viscosity model for CO₂ capture using hydroxypyridine-based ionic liquids. *Phys. Chem. Chem. Phys.* 19, 1134–1142.
- Boot-Handford, M.E., Abanades, J.C., Anthony, E.J., Blunt, M.J., Brandani, S., Mac Dowell, N., Fernandez, J.R., Ferrarri, M.C., Gross, R., Hallett, J.P., Haszeldine, R.S., Heptonstall, P., Lyngfelt, A., Makuch, Z., Mangano, E., Porter, R.T.J., Pourkashanian, M., Rochelle, G.T., Shah, N., Yao, J.G., Fennell, P.S., 2014. Carbon capture and storage update. *Energy Environ. Sci.* 7, 130–189.
- Brennecke, J.E., Gurkan, B.E., 2010. Ionic liquids for CO₂ capture and emission reduction. *J. Phys. Chem. Lett.* 1, 3459–3464.
- Cheng, J., Li, Y., Hu, L., Zhou, J., Cen, K., 2016. CO₂ adsorption performance of ionic liquid [P₆₆₆₁₄][2-Op] loaded onto molecular sieve MCM-41 compared to pure ionic liquid in biohythane/pure CO₂ atmospheres. *Energy Fuels* 30, 3251–3256.
- Cheng, J., Li, Y., Hu, L., Liu, J., Zhou, J., Cen, K., 2018. CO₂ absorption and diffusion in ionic liquid [P₆₆₆₁₄][Triz] modified molecular sieves SBA-15 with various pore lengths. *Fuel Process. Technol.* 172, 216–224.
- Chu, S., 2009. Carbon capture and sequestration. *Science* 325, 1599.
- Coasne, B., Viau, L., Vioux, A., 2011. Loading-controlled stiffening in nanoconfined ionic liquids. *J. Phys. Chem. Lett.* 2, 1150–1154.
- Cooper, A.L., 2015. Cooperative carbon capture. *Nature* 519, 294.
- Cox, P.M., Betts, R.A., Jones, C.D., Spall, S.A., Totterdell, I.J., 2000. Acceleration of global warming due to carbon-cycle feedbacks in a coupled climate model. *Nature* 408, 184–187.
- Cui, G., Wang, J., Zhang, S., 2016. Active chemisorption sites in functionalized ionic liquids for carbon capture. *Chem. Soc. Rev.* 45, 4307–4339.
- Drese, J.H., Choi, S., Lively, R.P., Koros, W.J., Fauth, D.J., Gray, M.L., Jones, C.W., 2009. Synthesis-Structure-Property relationships for hyperbranched aminosilica CO₂ adsorbents. *Adv. Funct. Mater.* 19, 3821–3832.
- Grosso, D., Boissière, C., Sanchez, C., 2007. Ultralow-dielectric-constant optical thin films built from magnesium oxyfluoride vesicle-like hollow nanoparticles. *Nat. Mater.* 6, 572.
- Gurkan, B.E., de la Fuente, J.C., Mindrup, E.M., Ficke, L.E., Goodrich, B.F., Price, E.A., Schneider, W.F., Brennecke, J.F., 2010. Equimolar CO₂ absorption by anion-functionalized ionic liquids. *J. Am. Chem. Soc.* 132, 2116–2117.
- Haszeldine, R.S., 2009. Carbon capture and storage: how green can black be? *Science* 325, 1647–1652.
- Hiremath, V., Jadhav, A.H., Lee, H., Kwon, S., Seo, J.G., 2016. Highly reversible CO₂ capture using amino acid functionalized ionic liquids immobilized on mesoporous silica. *Chem. Eng. J.* 287, 602–617.
- Horowitz, A.L., Panzer, M.J., 2014. Poly(dimethylsiloxane)-supported ionogels with a high ionic liquid loading. *Angew. Chem.-Int. Edit.* 53, 9780–9783.
- Jiang, D., Gao, J., Yang, J., Su, W., Yang, Q., Li, C., 2007. Mesoporous ethane-silicas functionalized with trans-(1R,2R)-diaminocyclohexane: relation between structure and catalytic properties in asymmetric transfer hydrogenation. *Microporous Mesoporous Mater.* 105, 204–210.
- Johari, K., Saman, N., Mat, H., 2014. Synthesis and characterization of novel sulfur-functionalized silica gels as mercury adsorbents. *J. Mater. Eng. Perform.* 23, 809–818.
- Kresge, C.T., Leonowicz, M.E., Roth, W.J., Vartuli, J.C., Beck, J.S., 1992. Ordered mesoporous molecular sieves synthesized by a liquid-crystal template mechanism. *Nature* 359, 710–712.
- Li, S., Wang, Z., Yu, X., Wang, J., Wang, S., 2012. High-performance membranes with multi-permeability for CO₂ separation. *Adv. Mater.* 24, 3196–3200.
- Li, K., Jiang, J., Yan, F., Tian, S., Chen, X., 2014. The influence of polyethyleneimine type and molecular weight on the CO₂ capture performance of PEI-nano silica adsorbents. *Appl. Energy* 136, 750–755.
- Liu, Z., Pudasainee, D., Liu, Q., Gupta, R., 2015. Post-combustion CO₂ capture using polyethyleneimine impregnated mesoporous cellular foams. *Sep. Purif. Technol.* 156, 259–268.
- Luo, X., Guo, Y., Ding, F., Zhao, H., Cui, G., Li, H., Wang, C., 2014. Significant improvements in CO₂ capture by pyridine-containing anion-functionalized ionic liquids

- through multiple-site cooperative interactions. *Angew. Chem.-Int. Edit.* 53, 7053–7057.
- Mello, M.R., Phanon, D., Silveira, G.Q., Llewellyn, P.L., Ronconi, C.M., 2011. Amine-modified MCM-41 mesoporous silica for carbon dioxide capture. *Microporous Mesoporous Mater.* 143, 174–179.
- Qi, G., Fu, L., Choi, B.H., Giannelis, E.P., 2012. Efficient CO₂ sorbents based on silica foam with ultra-large mesopores. *Energy Environ. Sci.* 5, 7368–7375.
- Ramdin, M., de Loos, T.W., Vlucht, T.J.H., 2012. State-of-the-art of CO₂ capture with ionic liquids. *Ind. Eng. Chem. Res.* 51, 8149–8177.
- Rochelle, G.T., 2009. Amine scrubbing for CO₂ capture. *Science* 325, 1652–1654.
- Sanz-Perez, E.S., Olivares-Marin, M., Arencibia, A., Sanz, R., Calleja, G., Maroto-Valer, M.M., 2013. CO₂ adsorption performance of amino-functionalized SBA-15 under post-combustion conditions. *Int. J. Greenh. Gas Control.* 17, 366–375.
- Sanz-Perez, E.S., Arencibia, A., Calleja, G., Sanz, R., 2018. Tuning the textural properties of HMS mesoporous silica. Functionalization towards CO₂ adsorption. *Microporous Mesoporous Mater.* 260, 235–244.
- Sayari, A., Liu, Q., Mishra, P., 2016. Enhanced adsorption efficiency through materials design for direct air capture over supported polyethylenimine. *ChemSusChem* 9, 2796–2803.
- Singh, M.P., Singh, R.K., Chandra, S., 2014. Ionic liquids confined in porous matrices: physicochemical properties and applications. *Prog. Mater. Sci.* 64, 73–120.
- Spigarelli, B.P., Kawatra, S.K., 2013. Opportunities and challenges in carbon dioxide capture. *J. CO₂ Util.* 1, 69–87.
- Tome, L.C., Marrucho, I.M., 2016. Ionic liquid-based materials: a platform to design engineered CO₂ separation membranes. *Chem. Soc. Rev.* 45, 2785–2824.
- Wan, M.M., Zhu, H.Y., Li, Y.Y., Ma, J., Liu, S., Zhu, J.H., 2014. Novel CO₂-capture derived from the basic ionic liquids orientated on mesoporous materials. *ACS Appl. Mater. Interfaces* 6, 12947–12955.
- Wan, Y., Zhao, 2007. On the controllable soft-templating approach to mesoporous silicates. *Chem. Rev.* 107, 2821–2860.
- Wang, C., Luo, H., Jiang, D.-e., Li, H., Dai, S., 2010. Carbon dioxide capture by superbase-derived protic ionic liquids. *Angew. Chem. Int. Ed.* 49, 5978–5981.
- Wang, C., Luo, H., Li, H., Zhu, X., Yu, B., Dai, S., 2012. Tuning the physicochemical properties of diverse phenolic ionic liquids for equimolar CO₂ capture by the substituent on the snion. *Chem.: Eur. J.* 18, 2153–2160.
- Wang, C., Luo, X., Zhu, X., Cui, G., Jiang, D.-e., Deng, D., Li, H., Dai, S., 2013. The strategies for improving carbon dioxide chemisorption by functionalized ionic liquids. *RSC Adv.* 3, 15518–15527.
- Wang, J., Huang, L., Yang, R., Zhang, Z., Wu, J., Gao, Y., Wang, Q., O'Hare, D., Zhong, Z., 2014. Recent advances in solid sorbents for CO₂ capture and new development trends. *Energy Environ. Sci.* 7, 3478–3518.
- Wang, L., Yao, M., Hu, X., Hu, G., Lu, J., Luo, M., Fan, M., 2015. Amine-modified ordered mesoporous silica: the effect of pore size on CO₂ capture performance. *Appl. Surf. Sci.* 324, 286–292.
- Xue, C., Tu, B., Zhao, D., 2008. Evaporation-induced coating and self-assembly of ordered mesoporous carbon-silica composite monoliths with macroporous architecture on polyurethane foams. *Adv. Funct. Mater.* 18, 3914–3921.
- Yan, X., Zhang, L., Zhang, Y., Qiao, K., Yan, Z., Komarneni, S., 2011. Amine-modified mesocellular silica foams for CO₂ capture. *Chem. Eng. J.* 168, 918–924.
- Yang, Y., Zhao, M., Yao, P., Huang, Y., Dai, Z., Yuan, H., Ni, C., 2018. Comparative studies on enzyme activity of immobilized horseradish peroxidase in silica nanomaterials with three different shapes and methoxychlor degradation of vesicle-like mesoporous SiO₂ as carrier. *J. Nanosci. Nanotechnol.* 18, 2971–2978.
- Zhang, S.J., Chen, Y.H., Li, F.W., Lu, X.M., Dai, W.B., Mori, R., 2006. Fixation and conversion of CO₂ using ionic liquids. *Catal. Today* 115, 61–69.
- Zhang, L.X., Li, P.C., Liu, X.H., Du, L.W., Wang, E.K., 2007. The effect of template phase on the structures of as-synthesized silica nanoparticles with fragile didodecyltrimethylammonium bromide vesicles as templates. *Adv. Mater.* 19, 4279–4283.
- Zhang, J., Sun, J., Zhang, X., Zhao, Y., Zhang, S., 2011. The recent development of CO₂ fixation and conversion by ionic liquid. *Greenh. Gases-Sci. Technol.* 1, 142–159.
- Zhang, J., Jia, C., Dong, H., Wang, J., Zhang, X., Zhang, S., 2013. A novel dual amino-functionalized cation-tethered ionic liquid for CO₂ capture. *Ind. Eng. Chem. Res.* 52, 5835–5841.
- Zhang, H., Goepfert, A., Czaun, M., Prakash, G.K.S., Olah, G.A., 2014. CO₂ capture on easily regenerable hybrid adsorbents based on polyamines and mesocellular silica foam. Effect of pore volume of the support and polyamine molecular weight. *RSC Adv.* 4, 19403–19417.
- Zhao, D., Feng, J., Huo, Q., Melosh, N., Fredrickson, G.H., Chmelka, B.F., Stucky, G.D., 1998. Triblock copolymer syntheses of mesoporous silica with periodic 50 to 300 angstrom pores. *Science* 279, 548–552.